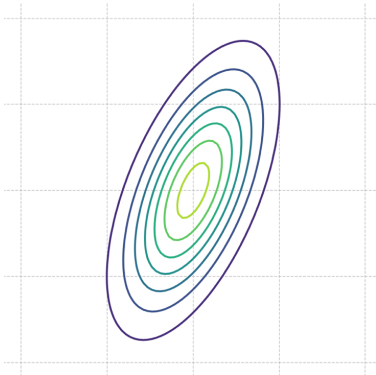

DSC 140A - Homework 08

Homework 08 Instructions. You may collaborate with whomever, including AI. Each student must submit their own homework even if they worked with others. We will not be grading for correctness, but we will be grading for completeness. Please turn out a single PDF file that contains your solutions to all of the problems. We recommend using $\text{\LaTeX}$ with our template. You can use markdown but make sure each problem is clearly labeled. For coding problems, we recommend using Jupyter notebooks, convert them into PDF, and then merge the PDF with your math problems solutions generated in $\text{\LaTeX}$.

Problem 1.

The picture below shows the contours of a multivariate Gaussian density function:



Which one of the following could possibly be the covariance matrix of this Gaussian?

- ☐ $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$
- ☐ $\begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}$
- ☐ $\begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix}$
- ☐ $\begin{pmatrix} 3 & -1 \\ -1 & 3 \end{pmatrix}$
- ☐ $\begin{pmatrix} 3 & -1 \\ -1 & 1 \end{pmatrix}$

Solution: C.

Problem 2.

Suppose a univariate Gaussian density function $\hat{f}$ is fit to a set of data using the method of maximum likelihood estimation (MLE).

True or False: $\hat{f}$ must be between 0 and 1 everywhere. That is, it must be the case that for every $x \in \mathbb{R}$, $0 < \hat{f}(x) \leq 1$. Write out your reasons.

Solution: False.

Video explanation: <https://youtu.be/zvpLrG4FYEc>

Problem 3.

Suppose a data set of 100 points in 10 dimensions is used in a binary classification task; that is, each point is labeled as either a 1 or a 0.

If Gaussian Naive Bayes is trained on this data, how many univariate Gaussians will be fit?

Solution: 20

Problem 4.

Let f_1 be a univariate Gaussian density with parameters μ and σ_1 . And let f_2 be a univariate Gaussian density with parameters μ and $\sigma_2 \neq \sigma_1$. That is, f_2 is centered at the same place as f_1 , but with a different variance.

Consider the density $f(x) = \frac{1}{2}(f_1(x) + f_2(x))$; the factor of $1/2$ is a normalization factor which ensures that f integrates to one.

True or False: f must also be a Gaussian density. Write out your reasons.

Solution: False. The sum of two Gaussian densities is not necessarily a Gaussian density, even if the two Gaussians have the same mean.

If you try adding two Gaussian densities with different variances, you will get:

$$f(x) = \frac{1}{2} \left(\frac{1}{\sqrt{2\pi}\sigma_1} e^{-\frac{(x-\mu)^2}{2\sigma_1^2}} + \frac{1}{\sqrt{2\pi}\sigma_2} e^{-\frac{(x-\mu)^2}{2\sigma_2^2}} \right)$$

For this to be a Gaussian, we'd need to be able to write it in the form:

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}},$$

but this is not possible when $\sigma_1 \neq \sigma_2$.

Problem 5.

The file at the link below contains a data set of 100 points from two classes (1 and -1).

https://f000.backblazeb2.com/file/jeldridge-data/003-two_clusters/data.csv

The first two columns contains features, and the last column contains the label of the point. Note that the labels are 1 and -1, not 1 and 0, and that there are no column headers.

In all parts of this problem you may use code to compute your answers. If you do, be sure to include your code.

- a) Suppose two Gaussians with full covariance matrices are used to model the densities $p_X(x|Y=1)$ and $p_X(x|Y=-1)$. What are the maximum likelihood estimates for the covariance matrices of each Gaussian?

(Allow each Gaussian to have its own covariance matrix; don't use the same covariance for both.)

Hint: the covariance matrix for the Gaussian fit to points from class 1 should have 12.29 in its top-left entry.

Solution: If you'd like to compute the covariances manually:

```
data = np.loadtxt('data.csv', delimiter=',')
X = data[:, :2]
y = data[:, -1]
```

```
X_1 = X[y == 1]
X_0 = X[y == -1]
```

```
y_1 = y[y == 1]
y_0 = y[y == -1] * 0
```

```
mu_1 = X_1.mean(axis=0)
Z_1 = (X_1 - mu_1)
n_1 = len(Z_1)
```

```
mu_0 = X_0.mean(axis=0)
Z_0 = (X_0 - mu_0)
n_0 = len(Z_0)
```

```
C_1 = 1 / n_1 * Z_1.T @ Z_1
C_0 = 1 / n_0 * Z_0.T @ Z_0
```

```
>>> C_1
array([[12.29584016,  0.28098224],
       [ 0.28098224, 16.06766736]])
>>> C_0
array([[10.91736224,  0.53015728],
       [ 0.53015728, 15.17320916]])
```

You could also have used numpy. If you did, you had to make sure to get to pass it an array whose *columns* are data points, instead of rows, and to use `bias = True`:

```
np.cov(X_1.T, bias=True)
```

- b) Using the estimated Gaussians with the Bayes classification rule, what are the predicted labels of each of the following points?

- $(0, 0)^T$
- $(1, 1)^T$
- $(10, 5)^T$
- $(5, -5)^T$
- $(8, 5)^T$

Show your calculations.

Note: making predictions in this way (using Gaussians with unequal covariance matrices) is known as *Quadratic Discriminant Analysis*.

Solution: The predicted labels are -1, -1, 1, -1, -1.

The code below defines a function for making the predictions:

```
mvn = scipy.stats.multivariate_normal
```

```
py_1 = n_1 / (n_1 + n_0)
py_0 = n_0 / (n_1 + n_0)
```

```
def predict(x):
    if mvn.pdf(x, mean=mu_1, cov=C_1) * py_1 > mvn.pdf(x, mean=mu_0, cov=C_0) * py_0:
        return 1
    else:
        return -1
```

Problem 6.

In lecture, we derived Linear Discriminant Analysis (LDA) by starting with the Bayes classifier and modeling each class-conditional density as a multivariate Gaussian and using the same covariance matrix for each. We stated, but did not prove, that the decision boundary of an LDA classifier is linear.

Recall that, for a binary classifier based on the Bayes Classifier, the decision boundary is the set of all points $\vec{x}$ where

$$\hat{p}(\vec{x} | Y = 1) \hat{\mathbb{P}}(Y = 1) = \hat{p}(\vec{x} | Y = 0) \hat{\mathbb{P}}(Y = 0),$$

where the various $\hat{p}$ and $\hat{\mathbb{P}}$ are estimated densities and probabilities.

Using this fact, prove that the decision boundary of an LDA classifier is linear. For simplicity, you may assume that $\vec{x} \in \mathbb{R}^2$ and that the shared covariance matrix is diagonal (although the result holds even if the covariance matrix is not diagonal).

Hint: since you may assume that $\vec{x} = (x_1, x_2)^T$, you can start from the above equality and solve for x_2 in terms of x_1 , showing that you get the equation of a line.

Solution: Let C be the shared covariance matrix. The decision boundary consists of all $\vec{x} = (x_1, x_2)^T$ for which

$$\hat{p}(\vec{x} | Y = 1) \cdot \hat{\mathbb{P}}(Y = 1) = \hat{p}(\vec{x} | Y = 0) \hat{\mathbb{P}}(Y = 0).$$

Since this is an LDA classifier, $p(\vec{x} | Y = 0)$ and $p(\vec{x} | Y = 1)$ are both multivariate Gaussian densities with the same covariance matrix C . Let $\vec{\mu}_0$ and $\vec{\mu}_1$ be the means of the two classes. Then, substituting the Gaussian density function, the decision boundary consists of all $\vec{x}$ for which

$$\frac{1}{(2\pi|C|)^{d/2}} e^{-\frac{1}{2}(\vec{x}-\vec{\mu}_0)^T C^{-1}(\vec{x}-\vec{\mu}_0)} \cdot \hat{\mathbb{P}}(Y = 0) = \frac{1}{(2\pi|C|)^{d/2}} e^{-\frac{1}{2}(\vec{x}-\vec{\mu}_1)^T C^{-1}(\vec{x}-\vec{\mu}_1)} \cdot \hat{\mathbb{P}}(Y = 1).$$

We notice that the same normalization term appears on both sides of the equation, and so we can cancel it:

$$e^{-\frac{1}{2}(\vec{x}-\vec{\mu}_0)^T C^{-1}(\vec{x}-\vec{\mu}_0)} \cdot \hat{\mathbb{P}}(Y = 0) = e^{-\frac{1}{2}(\vec{x}-\vec{\mu}_1)^T C^{-1}(\vec{x}-\vec{\mu}_1)} \cdot \hat{\mathbb{P}}(Y = 1).$$

We can also take the natural logarithm of both sides to simplify, giving:

$$-\frac{1}{2}(\vec{x} - \vec{\mu}_0)^T C^{-1}(\vec{x} - \vec{\mu}_0) + \log \hat{\mathbb{P}}(Y = 0) = -\frac{1}{2}(\vec{x} - \vec{\mu}_1)^T C^{-1}(\vec{x} - \vec{\mu}_1) + \log \hat{\mathbb{P}}(Y = 1). \quad (1)$$

Here is where it might be convenient to write $\vec{x} = (x_1, x_2)^T$ and expand. Additionally, write $\vec{\mu}_0 = (\mu_{01}, \mu_{02})^T$ and $\vec{\mu}_1 = (\mu_{11}, \mu_{12})^T$, and let

$$C = \begin{pmatrix} \sigma_1^2 & 0 \\ 0 & \sigma_2^2 \end{pmatrix}.$$

Since C is diagonal, its inverse is simply:

$$C^{-1} = \begin{pmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{1}{\sigma_2^2} \end{pmatrix}$$

Using these variables, we can expand:

$$\begin{aligned}
(\vec{x} - \vec{\mu}_0)^T C^{-1} (\vec{x} - \vec{\mu}_0) &= (x_1 - \mu_{01} \quad x_2 - \mu_{02}) \begin{pmatrix} \frac{1}{\sigma_1^2} & 0 \\ 0 & \frac{1}{\sigma_2^2} \end{pmatrix} \begin{pmatrix} x_1 - \mu_{01} \\ x_2 - \mu_{02} \end{pmatrix} \\
&= (x_1 - \mu_{01} \quad x_2 - \mu_{02}) \begin{pmatrix} \frac{x_1 - \mu_{01}}{\sigma_1^2} \\ \frac{x_2 - \mu_{02}}{\sigma_2^2} \end{pmatrix} \\
&= \frac{(x_1 - \mu_{01})^2}{\sigma_1^2} + \frac{(x_2 - \mu_{02})^2}{\sigma_2^2}.
\end{aligned}$$

Similarly, carrying out the same expansion for the other term, we will get:

$$(\vec{x} - \vec{\mu}_1)^T C^{-1} (\vec{x} - \vec{\mu}_1) = \frac{(x_1 - \mu_{11})^2}{\sigma_1^2} + \frac{(x_2 - \mu_{12})^2}{\sigma_2^2}.$$

Plugging these expansions back into Equation (1), we get:

$$-\frac{1}{2} \left(\frac{(x_1 - \mu_{01})^2}{\sigma_1^2} + \frac{(x_2 - \mu_{02})^2}{\sigma_2^2} \right) + \log \hat{\mathbb{P}}(Y = 0) = -\frac{1}{2} \left(\frac{(x_1 - \mu_{11})^2}{\sigma_1^2} + \frac{(x_2 - \mu_{12})^2}{\sigma_2^2} \right) + \log \hat{\mathbb{P}}(Y = 1).$$

Since we wish to solve for x_2 in terms of x_1 , we expand the squares and simplify:

$$\begin{aligned}
&-\frac{1}{2} \left(\frac{x_1^2 - 2\mu_{01}x_1 + \mu_{01}^2}{\sigma_1^2} + \frac{x_2^2 - 2\mu_{02}x_2 + \mu_{02}^2}{\sigma_2^2} \right) + \log \hat{\mathbb{P}}(Y = 0) \\
&= -\frac{1}{2} \left(\frac{x_1^2 - 2\mu_{11}x_1 + \mu_{11}^2}{\sigma_1^2} + \frac{x_2^2 - 2\mu_{12}x_2 + \mu_{12}^2}{\sigma_2^2} \right) + \log \hat{\mathbb{P}}(Y = 1).
\end{aligned}$$

We recognize some terms that appear on both sides of the equation, namely x_1^2/σ_1^2 and x_2^2/σ_2^2 . We can cancel these terms:

$$\begin{aligned}
&-\frac{1}{2} \left(-\frac{2\mu_{01}x_1}{\sigma_1^2} + \frac{\mu_{01}^2}{\sigma_1^2} - \frac{2\mu_{02}x_2}{\sigma_2^2} + \frac{\mu_{02}^2}{\sigma_2^2} \right) + \log \hat{\mathbb{P}}(Y = 0) \\
&= -\frac{1}{2} \left(-\frac{2\mu_{11}x_1}{\sigma_1^2} + \frac{\mu_{11}^2}{\sigma_1^2} - \frac{2\mu_{12}x_2}{\sigma_2^2} + \frac{\mu_{12}^2}{\sigma_2^2} \right) + \log \hat{\mathbb{P}}(Y = 1).
\end{aligned}$$

We now try to solve for x_2 in terms of x_1 . We start by moving all of the terms involving x_2 to the left-hand side of the equation, and all of the other terms to the right-hand side:

$$\left(-\frac{\mu_{02}}{\sigma_2^2} + \frac{\mu_{12}}{\sigma_2^2} \right) x_2 = \left(-\frac{\mu_{01}}{\sigma_1^2} + \frac{\mu_{11}}{\sigma_1^2} \right) x_1 + \frac{\mu_{01}^2}{2\sigma_1^2} - \frac{\mu_{11}^2}{2\sigma_1^2} + \frac{\mu_{02}^2}{2\sigma_2^2} - \frac{\mu_{12}^2}{2\sigma_2^2} - \log \hat{\mathbb{P}}(Y = 0) + \log \hat{\mathbb{P}}(Y = 1).$$

This is a linear function of x_1 and x_2 , and so the decision boundary is linear. We can see this a little more clearly if we give names to the coefficients of x_1 and x_2 and the constants:

$$\underbrace{\left(-\frac{\mu_{02}}{\sigma_2^2} + \frac{\mu_{12}}{\sigma_2^2} \right)}_{\alpha} x_2 = \underbrace{\left(-\frac{\mu_{01}}{\sigma_1^2} + \frac{\mu_{11}}{\sigma_1^2} \right)}_{\beta} x_1 + \underbrace{\frac{\mu_{01}^2}{2\sigma_1^2} - \frac{\mu_{11}^2}{2\sigma_1^2} + \frac{\mu_{02}^2}{2\sigma_2^2} - \frac{\mu_{12}^2}{2\sigma_2^2} - \log \hat{\mathbb{P}}(Y = 0) + \log \hat{\mathbb{P}}(Y = 1)}_{\gamma}.$$

Therefore: $\alpha x_2 = \beta x_1 + \gamma$, or: $x_2 = \frac{\beta}{\alpha} x_1 + \frac{\gamma}{\alpha}$, which is the equation of a line.

Problem 7.

You've been hired by a generic online retailer named after a rainforest named after a river. Your job is to build a model to predict whether or not a particular item will sell. You are provided with a dataset of outcomes for a collection of products:

Brand	Price Range	Condition	Sold
A	High	Used	No
A	High	New	Yes
B	Low	New	Yes
C	Medium	New	Yes
B	Low	Used	No
A	High	New	No
C	High	Used	Yes
A	Medium	Used	Yes
B	Medium	Used	No
C	Low	New	No
B	Low	Used	Yes

Using a Naïve Bayes classifier and the data above, predict if a product with Brand = B, Price Range = Medium, Condition = Used will sell or not. Show your calculations.

Solution:

We start by calculating the class conditional probabilities:

$$P(\text{Brand} = B \mid \text{Sold} = \text{Yes}) = \frac{2}{6}$$

$$P(\text{Brand} = B \mid \text{Sold} = \text{No}) = \frac{2}{5}$$

$$P(\text{Price Range} = \text{Medium} \mid \text{Sold} = \text{Yes}) = \frac{2}{6}$$

$$P(\text{Price Range} = \text{Medium} \mid \text{Sold} = \text{No}) = \frac{1}{5}$$

$$P(\text{Condition} = \text{Used} \mid \text{Sold} = \text{Yes}) = \frac{3}{6}$$

$$P(\text{Condition} = \text{Used} \mid \text{Sold} = \text{No}) = \frac{3}{5}$$

The prior probabilities are $P(\text{Sold} = \text{Yes}) = 6/11$ and $P(\text{Sold} = \text{No}) = 5/11$. Therefore:

$$P(\text{Sold} = \text{Yes} \mid \text{Brand} = \text{B}, \text{Price Range} = \text{Medium}, \text{Condition} = \text{Used})$$

$$\propto \frac{2}{6} \cdot \frac{2}{6} \cdot \frac{3}{6} \cdot \frac{6}{11}$$

$$= \frac{72}{2376} \approx 0.03$$

$$P(\text{Sold} = \text{No} \mid \text{Brand} = \text{B}, \text{Price Range} = \text{Medium}, \text{Condition} = \text{Used})$$

$$\propto \frac{2}{5} \cdot \frac{1}{5} \cdot \frac{3}{5} \cdot \frac{5}{11}$$

$$= \frac{30}{1375} \approx 0.021$$

Because the former is larger, we predict that the product will be **sold**.